

作业纸

课程名称: _____

班级: _____

教学班: _____

姓名: _____

学号: _____

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$$(b) \quad x(t) = e^{-t} \cdot e^{j\pi t} u(t) \\ = e^{(j\pi - 1)t} u(t)$$

$$\Re\{x(t)\} = \Re\{e^{(j\pi - 1)t} u(t)\}$$

$$y(t) = \Re\left\{\frac{1}{6} \left[e^{(j\pi - 1)t} - e^{-t} \right] u(t)\right\} \\ = \Re\left\{\frac{1}{6} (1-j) \left[e^{t \cos \pi} + e^{t \sin \pi} - e^{-t} \right] u(t)\right\}$$

$$= \Re\left\{\frac{1}{6} e^{-t} (\cos \pi t + j \sin \pi t - e^{-t}) - j \cos \pi t + \sin \pi t + e^{-t} j\right\} u(t) \\ = \frac{1}{6} e^{-t} [\cos \pi t - e^{-t} + \sin \pi t] u(t)$$

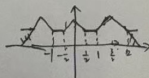
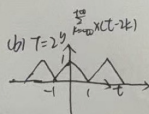
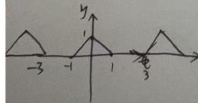
$$(a) \quad T=4$$

$$x(t) = \sum_{k=-\infty}^{\infty} \delta(t-4k)$$

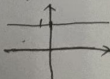
$$\Re\{y(t)\} = x(t) * \sum_{k=-\infty}^{\infty} \delta(t-4k)$$

$$= \sum_{k=-\infty}^{\infty} x(t) * \delta(t-4k) \quad (0) \quad T = \frac{3}{2} \sum_{k=-\infty}^{\infty} x(t - \frac{3}{2})$$

$$= \sum_{k=-\infty}^{\infty} x(t-4k)$$



$$(d) \quad \sum_{k=-\infty}^{\infty} x(t-1)$$



联系方式: _____

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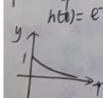
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$$\begin{aligned}
 2.4. \quad y[n] &= x[n] * h[n] \\
 &= \sum_{k=-\infty}^{+\infty} x[k] h[n-k] \\
 n \in [0, 7] \quad y[n] &= 0 \\
 7 \leq n \leq 12 \quad y[n] &= n-6 \\
 n \in [13, 23] \quad y[n] &= 6 \\
 n \in [24, 33] \quad y[n] &= 24-n \\
 n > 33 \quad y[n] &= 0
 \end{aligned}$$

$$\begin{aligned}
 12 \leq n \leq 13 \quad y[n] &= 6 \\
 n \in [13, 23] \quad y[n] &= 6 \\
 n \in [24, 33] \quad y[n] &= 24-n \\
 n > 33 \quad y[n] &= 0
 \end{aligned}$$

$$\begin{aligned}
 2.11 \quad (a) \quad x(t) &= u(t-3) - u(t-5) \\
 y(t) &= e^{-\pi t} u(t)
 \end{aligned}$$



$$\begin{aligned}
 3 < t < 5 \\
 y(t) &= \int_0^t e^{-\pi(t-\tau)} d\tau \\
 &= \frac{1}{\pi} (1 - e^{-\pi t})
 \end{aligned}$$

$$\therefore h(t-\tau) = e^{-\pi(t-\tau)} u(t-\tau)$$

$$\therefore y(t) = x(t) * h(t)$$

$$t \leq 3 \quad y(t) = 0$$

$$\begin{aligned}
 3 < t < 5 \\
 y(t) &= \int_0^t e^{-\pi(t-\tau)} d\tau \\
 &= \frac{1}{\pi} (1 - e^{-\pi t})
 \end{aligned}$$

联系方式: _____

$$\begin{aligned}
 5 < t < 7 \\
 y(t) &= \int_3^t e^{-\pi(t-\tau)} d\tau \\
 &= \frac{1}{\pi} (e^{-\pi(t-3)} - e^{-\pi(t-7)})
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad d(x(t) * y(t)) &= (x(t) * y(t))' \\
 \therefore y(t) &= h(t-3) - h(t-5) \\
 &= e^{-\pi(t-3)} u(t-3) - e^{-\pi(t-5)} u(t-5)
 \end{aligned}$$

$$(c) \quad y(t) = \frac{dy(t)}{dt}$$

$$2.17 \quad (a) \quad \frac{d}{dt} y(t) + 4y(t) = e^{-(1+j)t} u(t)$$

$$\begin{aligned}
 \therefore \gamma + 4 &= 0 \\
 \gamma &= -4 \quad y_h = C e^{-4t} u(t)
 \end{aligned}$$

$$\therefore y_p(t) = C e^{-(1+j)t} u(t)$$

$$\therefore y(t) = y_p(t) + y_h(t)$$

$$\therefore y_p(t) = k e^{-(1+j)t} u(t)$$

$$\therefore (1+j)k e^{-(1+j)t} = e^{-(1+j)t} u(t)$$

$$\therefore k = \frac{1}{1+j} = \frac{1-j}{2}$$

$$\therefore y_p = \frac{1-j}{2} e^{-(1+j)t} u(t)$$

$$\therefore y(t) = C e^{-4t} u(t) + \frac{1-j}{2} e^{-(1+j)t} u(t)$$

$$m, y(0) = 0 \quad C + \frac{1-j}{2} = 0$$

$$\therefore y(t) = \frac{1-j}{2} (e^{-(1+j)t} - e^{-4t}) u(t)$$